

Installing TrueNAS to Create an iSCSI Target

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TrueNAS is a software solution that allows you to transform a computer into a NAS (network-attached storage) server. There are two editions: full-featured paid TrueNAS Scale for enterprises (based on Debian Linux) and free TrueNAS Core (based on FreeBSD) formerly known as FreeNAS.

TrueNAS can be used to deploy shared storage for VMware ESXi hosts and Hyper-V servers. This blog post explains how to install TrueNAS Core and configure an iSCSI target that can be used in different scenarios, including shared storage for VMware vSphere.

Complete data protection for VMs and instant recovery options. Secure backup targets onsite, offsite and in the cloud. Anti-ransomware features.

TrueNAS Installation System Requirements

TrueNAS Core (FreeNAS) minimum hardware requirements are:

- CPU: 64-bit processor (Intel or AMD)
- RAM: 4 GB
- Storage: At least one additional disk drive for data storage
 - 1 x 10Gb for Nas Installation
 - 1 x 5Gb for NFS Service
 - 1 x 10Gb for iScsi Service
 - 1 x 8Gb for SMB/Cifs Service
- Network: One Ethernet ports

Notes on requirements:

- The requirements for installing TrueNAS Core that are listed above are general guidelines and the actual hardware needs may vary based on factors such as the size of your storage environment, the number of users, and the intended use case.
- TrueNAS is flexible and can run on various hardware configurations, including custom-built systems, servers, and purpose-built storage appliances.
- TrueNAS supports a variety of network interface cards (NICs), storage controllers, and hardware components. However, it's crucial to ensure compatibility, especially for components that require drivers in FreeBSD.
- The boot device is used to store the operating system, and using a reliable USB flash drive or SSD is recommended.
- For optimal performance and reliability, consider using ECC (error-correcting code) RAM, especially in environments where data integrity is critical.
- Always refer to the official TrueNAS documentation for the specific version you plan to install, as system requirements and recommendations may be updated with new releases.
- It is recommended that you use multiple disk drives to ensure storage redundancy.

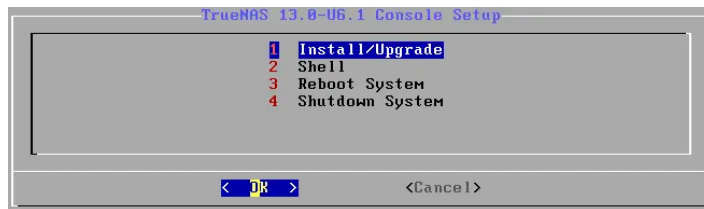
How to Install TrueNAS Core on a VM

Let's configure an iSCSI target on TrueNAS Core using the virtual environment. TrueNAS Core installed on a VM will be used to create an iSCSI target in this example.

TrueNAS Core is a free distribution based on the FreeBSD operating system, which provides a web interface for creating and managing network shares.

NOTE: The process of installing TrueNAS Core of configuration is identical to installing the previous version FreeNAS.

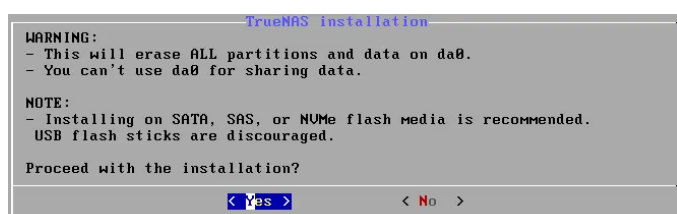
1. Open VMware Workstation and create a new VM with the following parameters:
 - 1 x 10Gb for Nas Installation
 - 1 x 5Gb for NFS Service
 - 1 x 10Gb for iScsi Service
 - 1 x 8Gb for SMB/Cifs Service
 - Network adapter: **Bridge**
 - Memory: **4 GB**
 - CPU: **1 processor**
 - The VM name: TrueNAS13-0
2. Power on the VM, boot from the TrueNAS Core ISO installation image, and **install TrueNAS Core** on the VM. The installation process is straightforward – answer the questions provided by the installation wizard.
3. Select **1 Install/Upgrade** and hit **OK** to proceed.



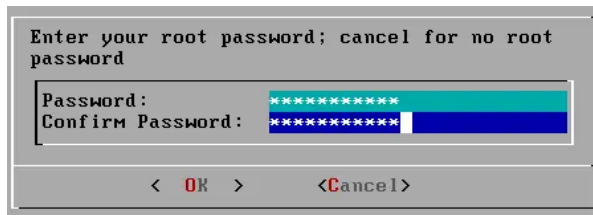
4. Select a virtual disk to install an operating system. Install TrueNAS Core on the 10-GB disk.



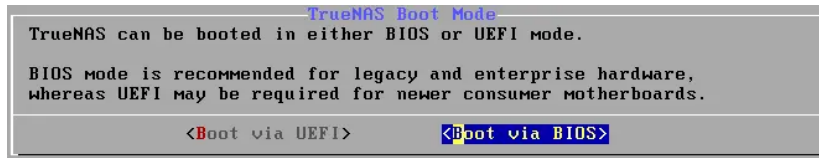
5. Read the warning and hit **Yes** to erase the selected disk and continue.



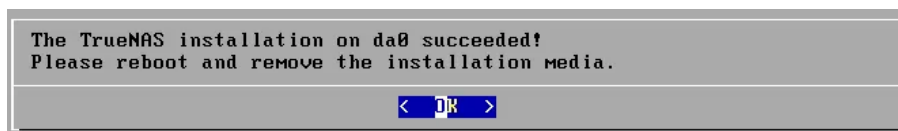
6. Enter and confirm the root password (Pass.1234). This password will also be used to log in to the web interface of TrueNAS Core. Remember the root password you specify.



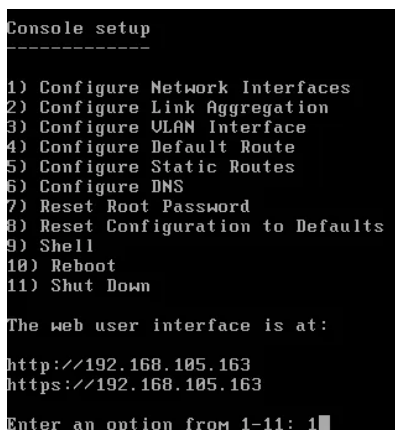
7. Select the BIOS or UEFI mode to use for operating system boot.



8. When you see the message that the TrueNAS installation has succeeded, hit OK and reboot the TrueNAS13-0 virtual machine.



9. Check the network settings of TrueNAS Core. The IP address is obtained via DHCP (192.168.105.134). Press **1** (Configure Network Interfaces) in the main menu and set a static IP address such as 192.168.105.105. The web server interface link is displayed in the TrueNAS console interface.



- Select an interface: **1**
- Remove the current settings of this interface? **n**
- Configure interface for DHCP? **n**
- Configure IPv4? **y**
- Interface name: **eth0**
- IPv4 address: **192.168.105.105**
- IPv4 netmask: **255.255.255.0**
- Configure IPv6? **n**

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11) Shut Down

The web user interface is at:
http://192.168.105.163
https://192.168.105.163

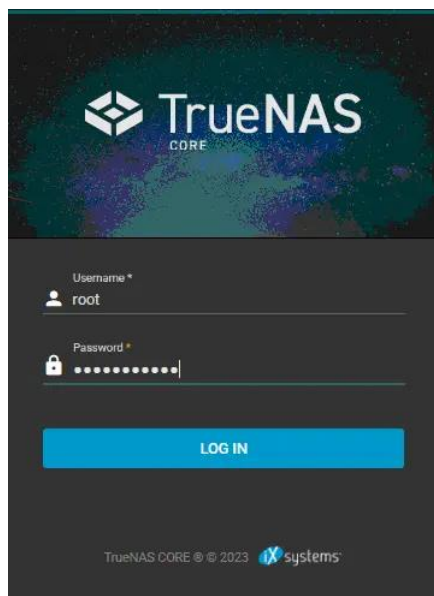
Enter an option from 1-11: 1
1) em0
Select an interface (q to quit): 1
Remove the current settings of this interface? (This causes a momentary disconnection of the network.) (y/n) n
Configure interface for DHCP? (y/n) n
Configure IPv4? (y/n) y
Interface name: eth0
Several input formats are supported
Example 1 CIDR Notation:
192.168.1.1/24
Example 2 IP and Netmask separate:
IP: 192.168.1.1
Netmask: 255.255.255.0, /24 or 24
IPv4 Address: 192.168.105.105
IPv4 Netmask: 255.255.255.0
Saving interface configuration: Ok
Configure IPv6? (y/n) n

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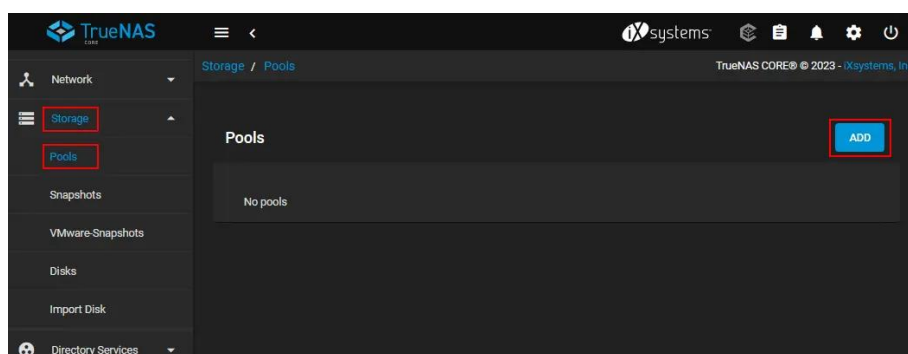
Set Up an iSCSI Target on TrueNAS Core

Once the TrueNAS installation has been finished and network settings have been configured, you can set up an iSCSI target on TrueNAS Core.

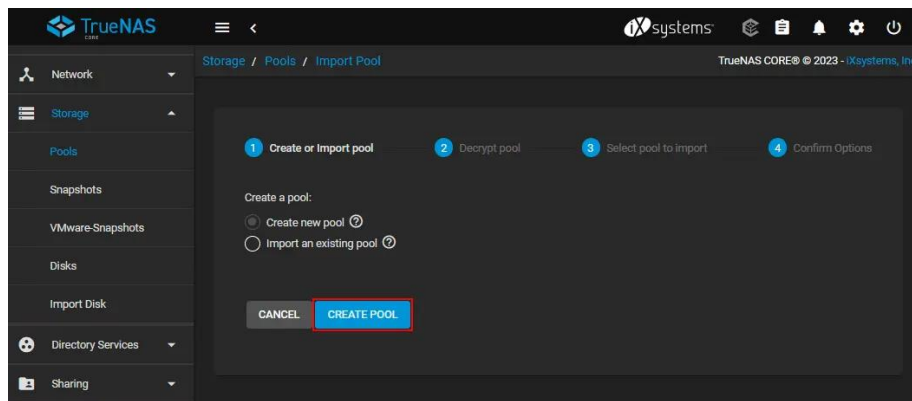
1. Enter the address of the TrueNAS Core web interface in your browser. In this example, <http://192.168.105.105> is the address. Then select the legacy web interface and enter the root username and the password you have specified during the TrueNAS installation. Click **Log in**.



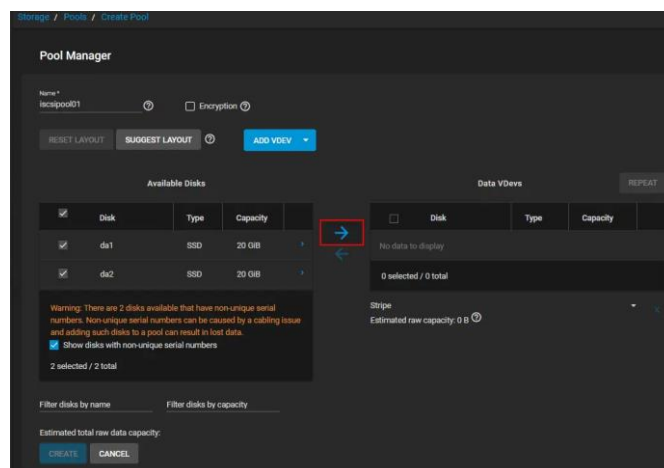
2. Click **Storage > Pools** and then click **Add** to create a new storage pool.



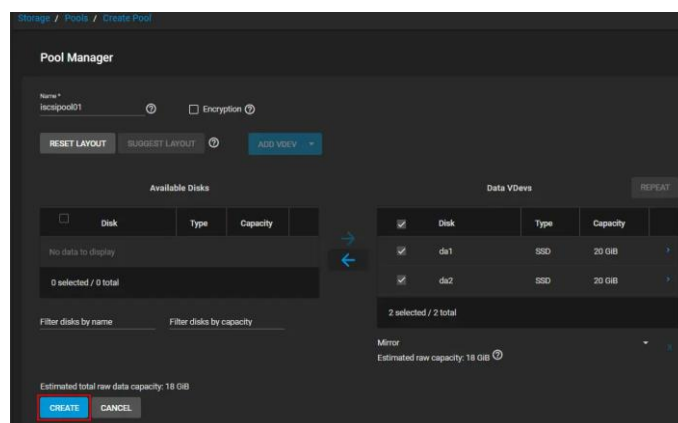
3. Select the **Create new pool** option and click **Create Pool**.



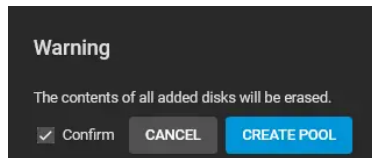
4. Configure a new pool:
 - Enter a pool name, for example, **iscsipool01**.
 - Select the two virtual disks, which were added to the TrueNAS VM configuration in the *Available disks* section. In our case, we use two 20-GB disks. Using two disks allows you to provide redundancy.
 - Click the right arrow → in the web interface to add these two selected disks to the pool.



- Make sure that the selected disks are displayed in the *Data VDevs* section.
- Click **Create**.

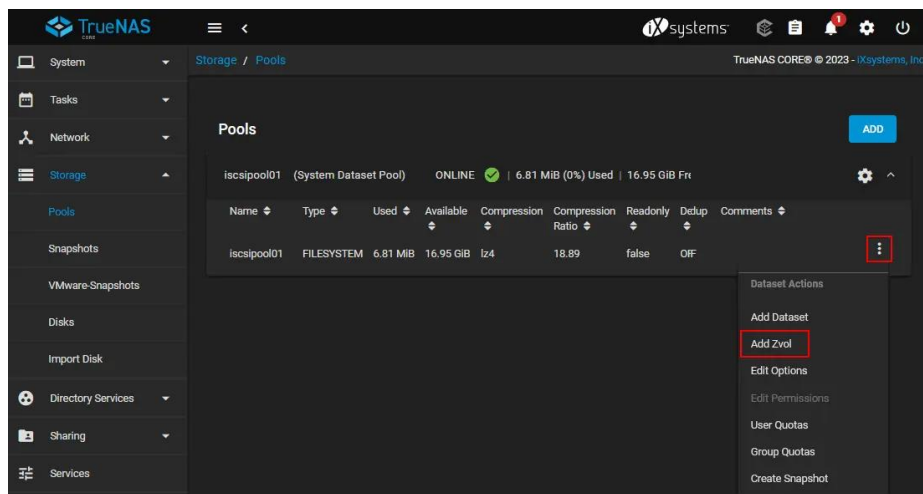


5. Read the warning message, select the **Confirm** checkbox to confirm the action, and click **Create Pool**.



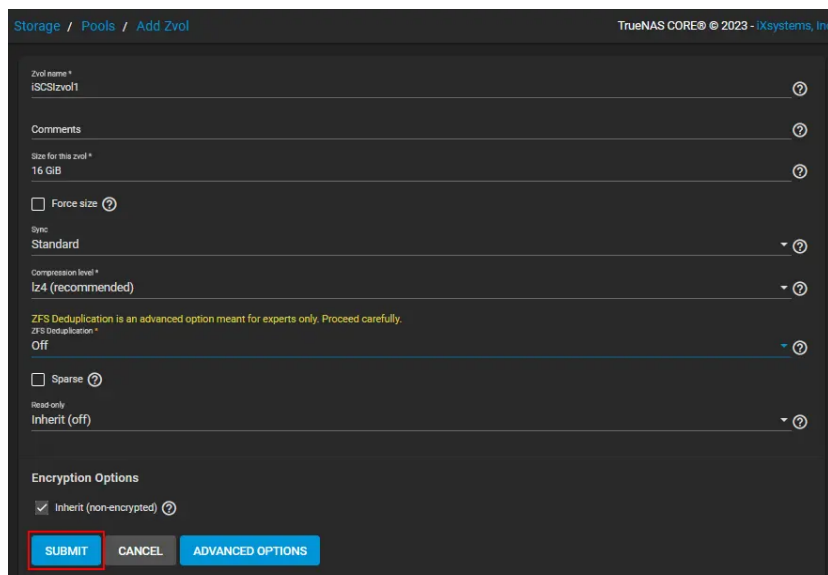
6. A new pool has been created in TrueNAS. Next, we need to add a *zvol*. This is a feature of the Zettabyte File System (ZFS) for creating a raw block device over ZFS that allows you to use a *zvol* as an iSCSI extent.

Click the **three dots** icon for your storage pool and hit **Add Zvol** in the menu that opens.

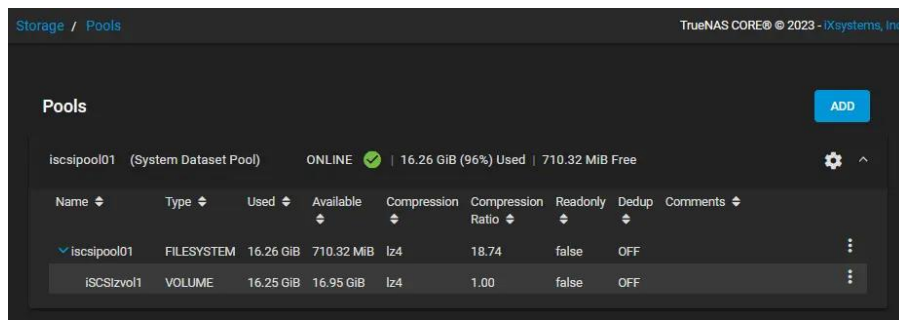


7. In this example, the configured parameters of a zvol are:
 - Zvol name: *iSCSIzvol1*
 - Size for this Zvol: *16 GB* (you should not use more than 80% of your available space for your zvol). 16 GB is equal to 80% of 20 GB.
 - Sync: *Inherit (standard)*
 - Compression level: *Inherit (lz4)*
 - ZFS deduplication: *Inherit (off)*

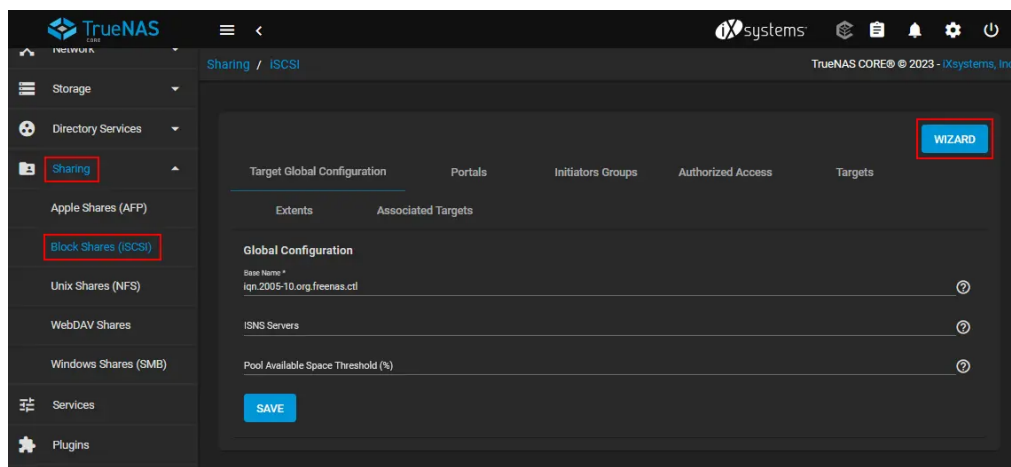
Click the **Save** button to create a new zvol.



8. A zvol has been created and is displayed under the pool in **Storage > Pools**.

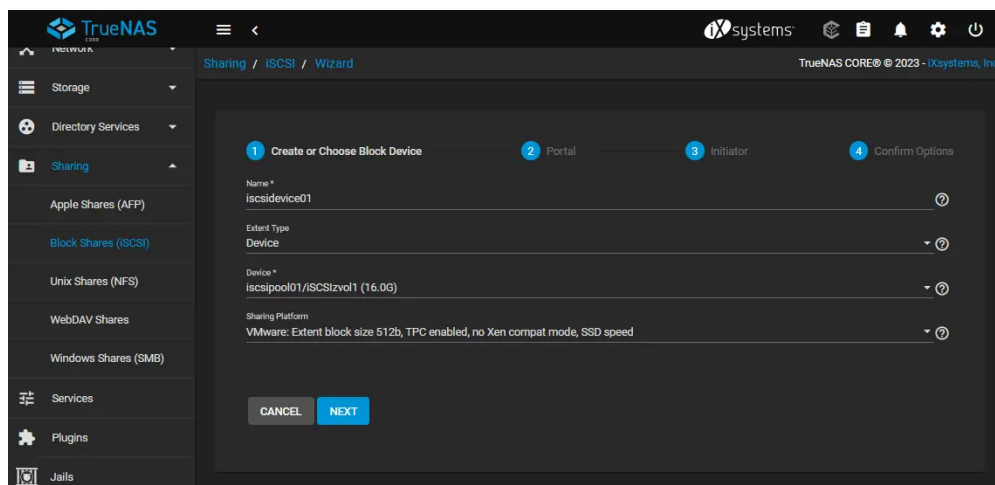


9. Go to **Sharing > Block Shares (iSCSI)** and click **Wizard** to create a new iSCSI share.



10. **Create or Choose Block Device.** We use the following parameters:
- Name: *iscsidevice01*
 - Type: *Device*
 - Device: *iscsipool01/iscsizvol01*
 - What are you using this for: *VMware: Extent block size 512b, TPC enabled, no Xen compat mode, SSD speed*

Click **Next** at each step of the wizard to continue.



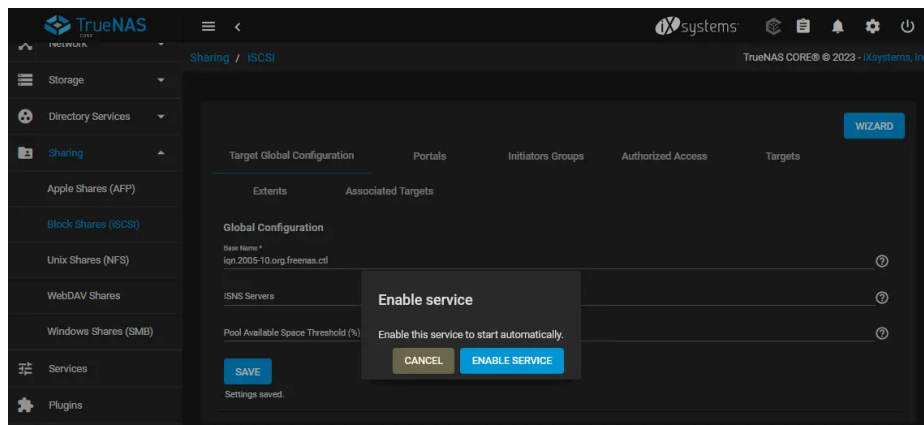
11. **Portal.** Set the needed parameters for a portal. In our example, the parameters are the following:
- Portal: *Create New*

- Discovery Auth Method: *NONE*
- Discovery Auth Group: *None*
- IP: *0.0.0.0* (Allows access from any IP address. If you want to use a more secure approach, you can define custom trusted IP addresses).
- Port: 3260 (is used by default)

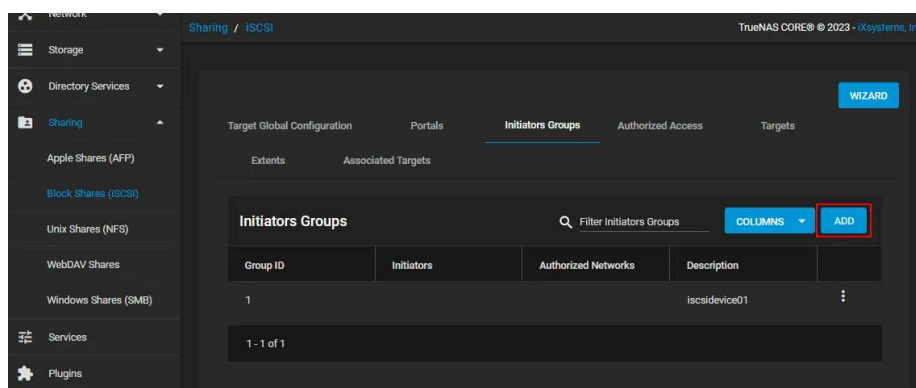
12. **Initiator.** At this step, you can configure access for iSCSI initiators. We leave the default values. Click the ? icon to read tips.

13. **Confirm Options.** Verify the configuration for your iSCSI target and click **Submit**.

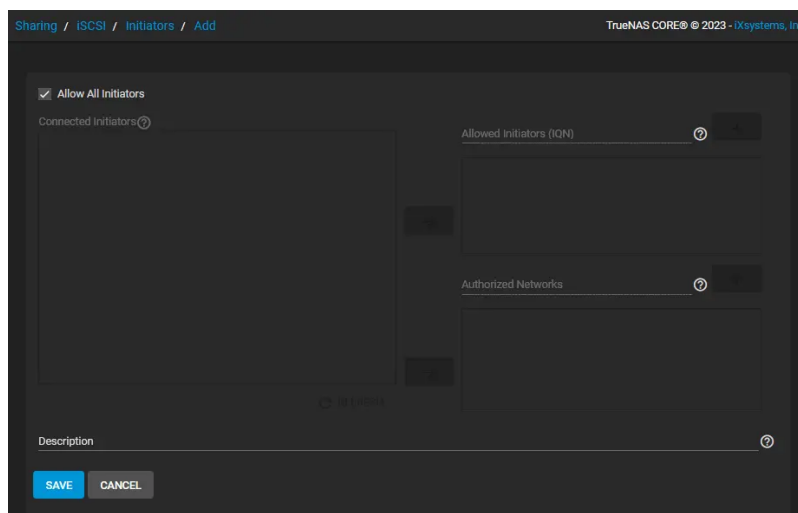
14. Click **Enable service** when you see a pop-up message.



15. You can edit settings for iSCSI shares if needed. On the **Sharing > Block Shares (iSCSI)** page, you can select the **Initiators** tab and click **Add**.



16. Select the **Allow Initiators** checkbox.



17. You can also edit the Extent settings if needed.

The screenshot shows the 'Basic Info' tab for an iSCSI extent. The 'Name' is 'iscsidvce01'. The 'Description' is empty. The 'Enabled' checkbox is checked. The 'Type' is 'Device'. The 'Device' is 'iscsipool01/iscsizvol1 (16.0G)'. The 'Serial' is '000c29e91076000'. The 'Logical Block Size' is '512'. The 'Disable Physical Block Size Reporting' checkbox is unchecked. The 'Available Space Threshold (%)' is empty. The 'Compatibility' section has 'Enable TPC' checked, 'Xen initiator compat mode' unchecked, and 'LUN RPM' set to 'SSD'. The 'Read only' checkbox is unchecked. There are 'SAVE' and 'CANCEL' buttons at the bottom.

18. Once the iSCSI target has been configured, you can connect this iSCSI target to ESXi hosts. Remember the base name of the iSCSI target that is displayed in the **Sharing > Block shares (iSCSI) > Target Global Configuration** tab. In this case, **iqn.2005-10.org.freenas.ctl** is the Base Name that would be used for connecting the iSCSI Target to a client as a shared datastore.

The screenshot shows the 'Target Global Configuration' tab for an iSCSI target. The 'Base Name' is 'iqn.2005-10.org.freenas.ctl'. The 'ISNS Servers' are empty. The 'Pool Available Space Threshold (%)' is empty. There is a 'SAVE' button at the bottom. The left sidebar shows the 'Sharing' menu expanded, with 'Block Shares (iSCSI)' selected. The top right shows the 'WIZARD' button.

Now, you can configure an iSCSI initiator on an host or another machine and connect to the iSCSI target configured on TrueNAS Core.

Conclusion

Installing TrueNAS Core is affordable and the installation process is straightforward. To configure an iSCSI target, you have to create a storage pool, create a zvol, add extents, and configure allowed IP addresses. The configuration is made in a user-friendly web interface. You can also use TrueNAS Core to configure file shares such as SMB and NFS and use them to store backups with NAKIVO Backup & Replication.